Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/20/2007
PI Name:	Beckermann, Christoph Ph.D.		
Project Title:	Microstructure Evolution in Free Dendritic Growth		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1527	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	10/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 10/31/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNM04AA18G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The tasks of this project are: (1) to study transient growth of multiple interacting equiaxed dendrites; (2) to study free growth of binary alloy alloy dendrites in the presence of coupled heat and solute transport. The tasks will be accomplished through a combination of experiments, using transparent model substances in an existing dendrite growth setup, and numerical simulations using the phase-field and other methods. The objective of the project is to study microstructural evolution in free dendritic growth of pure substances and alloys from a supercooled melt. The research will not only close important gaps in our current understanding of dendritic growth, but is also needed for		

	placing current simulations of grain structure evolution in castings on firmer theoretical grounds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study is important for improving the understanding and computer simulation of solidification processing of metals and alloys.
Task Progress:	During the no-cost extension, work was completed on two journal papers on the effect of convection on free dendritic growth. Experiments are conducted to measure the dendrite tip growth velocities of equiaxed crystals of the transparent model alloy succinonitrile-acetone that are settling in an undercooled melt. The tip velocities are measured as a function of the crystal settling speed and the Eulerian angle between the dendrite arms and the flow direction relative to the crystal. The ratio of the settling speed (or flow velocity) to the tip growth velocity ranges from 62 to 572. The ratio of the measured tip velocity to that predicted from a standard diffusion theory for free dendritic growth ranges from almost zero for dendrite tips growing in the wake of the crystal, about unity for dendrite tips with an orientation close to normal to the flow direction, to up to two for dendrite tips growing into the flow. Despite the relatively strong flow relative to the crystal, the average tip growth velocity of the six primary dendrite arms of an equiaxed crystal is found to be in excellent agreement with the standard diffusion theory result. The individual tip velocities are correlated using a boundary layer model of free dendritic growth in the presence of melt flow that is modified to account for the flow angle dependence. Using the same dendrite tip selection parameter as established previously under purely diffusive conditions (0.02), good agreement is achieved between the measured and predicted tip velocities. The model is also found to predict well the variations in the tip velocity that occur during settling due to crystal rotation and settling speed changes. Experiments are conducted to measure the internal solid fraction evolution of equiaxed dendritic crystals that are freely growing and settling in an undercooled melt using the transparent model alloy succinonitrile-acetone. The internal solid fraction is determined from the measured settling speed and crystal envelope shape and size. Depending on the melt undercooling and acetone concentration, the internal solid fraction is found to vary between 0.55 and 0.1. In all experiments, the internal solid fraction decreases continually during settling. Based on heat and solute balances, a model is developed for predicting the internal solid fraction evolution under convective conditions. Nusselt and Sherwood number correlations are obtained that allow for the calculation of the thermal and solutal boundary layer thicknesses at the crystal envelope. The measured and predicted internal solid fraction evolutions are found to be in good agreement.
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Articles in Peer-reviewed Journals	Badillo A, Ceynar DL, Beckermann C. "Growth of equiaxed dendritic crystals settling in an undercooled melt, Part 1: Tip kinetics." Journal of Crystal Growth. 2007 Dec 1;309(2):197-215. http://dx.doi.org/10.1016/j.jcrysgro.2007.09.018 , Dec-2007
Articles in Peer-reviewed Journals	Badillo A, Ceynar DL, Beckermann C. "Growth of equiaxed dendritic crystals settling in an undercooled melt, Part 2: Internal solid fraction." Journal of Crystal Growth. 2007 Dec 1;309(2):216-24. http://dx.doi.org/10.1016/j.jcrysgro.2007.09.019 , Dec-2007
Papers from Meeting Proceedings	Badillo A, Beckermann C. "Growth of Equiaxed Dendritic Crystals Settling in an Undercooled Melt." Solidification Processing 2007, Sheffield, UK, July 2007. In: Solidification Processing 2007, ed. by H. Jones, Dept. Engineering Materials. Sheffield, UK : The University of Sheffield, 2007, pp. 176-180. , Jul-2007